Anna Wang

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Education

Carnegie Mellon University — Pittsburgh, PA
B.S. in Electrical and Computer Engineering **GPA**: 3.71/4.00 — May 2029
Minor in Neural Computation

Relevant Courses: Effective Coding with AI, Structure and Design of Digital Systems, Introduction to Probability Theory, Principles of Imperative Computation, Signal and Systems

Skills

Programming: Python, SystemVerilog, C, MATLAB, Java, JavaScript
Spoken Languages: English, Mandarin (Chinese), French

Work Experience

Biomedical Functional Imaging and Neuroengineering Laboratory — Pittsburgh, PA
Brain-Computer Interface Undergraduate Research Assistant — Mar 2026 – Present

- Migrated lab's SpeechBCI experimental pipeline from BCI2000 to modular Lab Streaming Layer + PsychoPy stack to support new hardware.
- Designed and built all stimulus frontend logic for data collection sessions currently used in active human subject experiments to train an EEG-to-speech decoder algorithm.
- Contributed to an end-to-end SpeechBCI pipeline to decode EEG signals into natural language using deep learning sequence models.

MIT Lincoln Laboratory — Lexington, MA
Advanced Concepts & Technologies Intern (Group 39) — Jul 2024 – Aug 2024

- Built a 700 Hz Morse code communication system.
- Designed circuits and filters with LTSpice and MATLAB and analyzed signals through an oscilloscope and spectrum analyzer.

Duke Center for Computational Evolutionary Intelligence — Shrewsbury, MA
Research Intern — Jul 2022 – Jun 2025

- Built a 6-layer 1D convolution neural network (CNN) to classify sleep stages to detect sleep disorders achieving ~99% accuracy based on the EEG Sleep EDFX dataset.
- Quantized CNN to 2 bytes & 1 byte achieving ~60% and implemented Quantization-Aware Training.
- Published in IEEE Xplore: **A. Wang**, Q. Huang, Y. Chen, *A Quantized Parsimonious CNN Model for Sleep Polysomnogram Data Streams*.

Project Experience

Image Rendering Tiny GPU
18-240: Structure and Design of Digital Systems Final Project — Jul 2026 – Present

- Designed an 8-lane lockstep SIMD GPU in SystemVerilog with a custom RV32IM ISA, including a multi-cycle core, scheduler, ALU, control flow, and load/store support.
- Built an AI-assisted Python toolchain for assembly and image-to-memory conversion, plus a shared-memory arbiter for concurrent byte-masked accesses across all eight lanes.
- Rendered a user-selected image on a Xilinx Spartan-7 FPGA display and verified GPU through 10 testbenches, 801 assertions, and mutation testing.

Activities

15-112 Teaching Assistant, Carnegie Mellon University — Jul 2026 – Present
Club Tennis Events Chair, Carnegie Mellon University — Sept 2025 – Present
Asian Student Association, Carnegie Mellon University — Sept 2025 – Present